

An efficient Epidemic Routing Protocol with Reinforcement Learning algorithm in Opportunistic Networks

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Abstract—In many African countries, mobile payments are vital for financial transactions. However, enabling peer-to-peer payments is challenging due to limited network infrastructure and resource-constrained devices. Companies like Ejara are working to bring accessible financial services to underserved communities. However, existing routing protocols for Opportunistic Networks (OppNets), like traditional epidemic routing, create high routing overhead and latency from indiscriminate packet flooding, putting significant pressure on network and device resources. This study addresses these issues by introducing an optimized epidemic routing protocol with Reinforcement Learning algorithms integration to enhance packet forwarding in OppNets. The protocol dynamically adapts forwarding decisions based on delivery probability, latency, and resource constraints. Emulation results demonstrate that this approach significantly reduces routing overhead and latency while improving packet delivery reliability. This solution has meaningful implications for enabling efficient mobile payments and peer-to-peer interactions, especially in resource-constrained environments with intermittent connectivity, supporting broader accessibility to financial services.

Index Terms—Opportunistic Networks, Epidemic Routing, Reinforcement Learning, Delay Tolerant Networks, Mobility Models.

I. INTRODUCTION

In some traditional markets across many regions of Africa, where daily transactions play an important part of everyday life, there is a growing shift away from using physical cash. This shift is evidenced by the widespread adoption of mobile payments among internet users, with 84 percent in Kenya and 60 percent in Nigeria opting for digital transactions in 2021 [1]. Several African governments are promoting the use of mobile payments, especially in response to the challenges posed by the COVID-19 pandemic [2]. This transition to digital financial solutions plays a crucial role in promoting financial inclusion, particularly in regions where a large number of the population lacks access to traditional banking services.

However, the rapid growth of digital transactions in regions with limited network infrastructure brings new challenges, as many transactions rely on stable internet connectivity. This issue is especially prevalent in environments like traditional marketplaces across Africa, where network access is often limited or sporadic [3]. OppNets [4]—an approach within Delay-Tolerant Networks—are based on an established concept that can bring solutions for these use cases. OppNets enables

direct, device-to-device communication, it also allows secure, efficient transactions on mobile phones without a dependable internet connection, making them a promising solution for facilitating mobile payments in such areas [5].

By using OppNets, mobile payments can continue to function effectively in these challenging conditions, thereby addressing the growing demand for digital financial transactions in regions with unpredictable or limited network access. However, ensuring reliable message delivery in such environments presents its own set of challenges, especially when connectivity is highly dynamic and unpredictable.

To address connectivity challenges in resource-constrained environments, Epidemic Routing [6] has emerged as a potential solution, relying on a controlled broadcast of multiple message copies to ensure eventual delivery even without a stable path. While effective, this method can lead to excessive bandwidth usage, storage exhaustion, and rapid energy depletion on devices with limited resources, making it unsustainable in many scenarios. Reinforcement Learning (RL) [7] techniques offer an adaptive solution by allowing nodes to learn efficient routing strategies over time, reducing redundant message transmissions and conserving resources. Integrating RL algorithms into Epidemic Routing enhances message delivery success while minimizing resource consumption, addressing key challenges in low-infrastructure, resource-limited networks [8].

This paper introduces a novel integration of RL algorithms into Epidemic Routing to improve packet forwarding efficiency in OppNets. By leveraging RL algorithms, nodes can make informed decisions on when and to whom to forward messages. The goal is to prioritize forwarding to neighbors that are more likely to successfully deliver the message, thereby reducing overhead while maintaining a high delivery rate.

The main contributions of this paper are as follows:

- We model OppNets and introduce a RL-enhanced Epidemic Routing protocol. This protocol enables nodes to intelligently select forwarding neighbors by leveraging learned experiences, specifically optimizing message transmission in environments constrained by limited network infrastructure and computational resources.
- We evaluate the proposed protocol using the Mininet-WiFi emulator [9]¹ in various network scenarios, demonstrating

¹<https://github.com/intrig-unicamp/mininet-wifi>

its effectiveness in reducing message delivery delays and overhead compared to Traditional Epidemic Routing, while maintaining high reliability in resource-constrained environments.

The rest of this paper is structured as follows: Section II reviews related work on OppNets, Epidemic Routing and RL algorithm. Section III outlines the system overview and our implementation. In Section IV, we discuss the experimental setup and performance evaluation. Section V concludes the paper and provides insights into potential future work.

II. RELATED WORKS

In this section, we present background about OppNets and epidemic routing. We also investigate some emulation tools that have been used in Delay-Tolerant Network (DTN) research.

A. Opportunistic Network

OppNets with characteristics like dynamic and self-organizing nature, operating on temporary connections rather than fixed routes [5], enabling data exchange in challenging conditions like sparse infrastructure or variable topologies. These networks leverage intermittent connectivity among nodes, allowing communication even without direct pathways through intermediate nodes. OppNets address the need for communication in highly unpredictable environments by adapting to the unstable topology and intermittent connections present in such scenarios.

B. Epidemic Routing

Epidemic routing is a communication protocol designed for DTNs, characterized by intermittent connectivity and long delays [6] [10]. Inspired by epidemiological models, this protocol disseminates messages through random exchanges among nodes, ensuring that all nodes eventually receive all messages. Each message is stored with a unique identifier, and when two nodes connect, they share their message inventories to synchronize their data. While epidemic routing is robust and simple to implement, it can consume significant resources and requires effective buffer management to optimize performance.

C. Reinforcement Learning in Opportunistic Networks

Reinforcement learning [7] is a machine learning paradigm where agents learn to make decisions by interacting with their environment, optimizing actions to maximize a cumulative reward. In the context of OppNets, RL offers a promising approach to improve message forwarding efficiency by enabling nodes to dynamically adapt their forwarding decisions based on network conditions [11]. Given the intermittent connectivity and highly variable topology of OppNets, traditional protocols often struggle with high overhead and inefficiencies. RL can address these challenges by allowing nodes to learn optimal forwarding strategies over time, improving delivery rates and reducing resource consumption.

III. EPIDEMIC ROUTING PROTOCOL WITH REINFORCEMENT LEARNING ALGORITHMS

A. System Overview

In our innovative approach to RL integration with OppNets, we introduce a system model as illustrated in Fig. 1. In this

study, we model OppNets where mobile nodes move freely and intermittently encounter one another. Each node is equipped with a wireless interface, allowing it to communicate directly with nearby nodes within its transmission range. The nodes lack any fixed infrastructure, making routing and data dissemination highly dynamic and dependent on node mobility patterns and encounters.

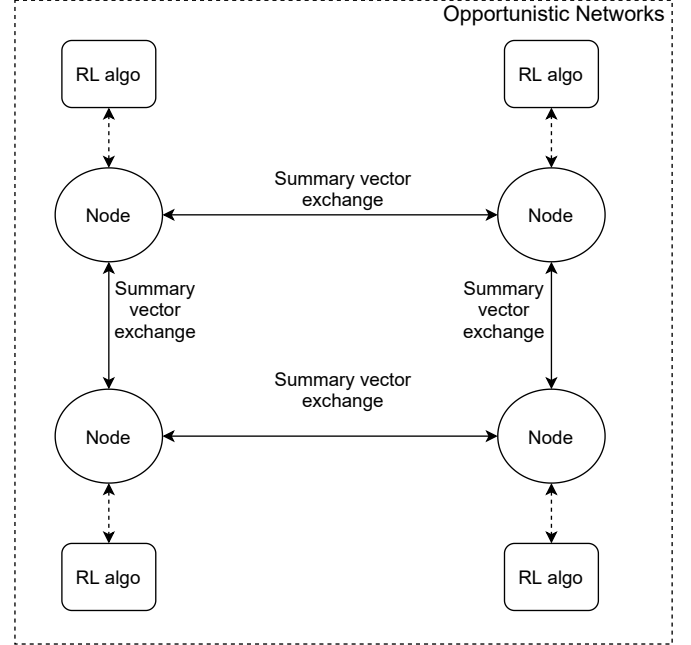


Fig. 1: Overview of Opportunistic Networks Utilizing Epidemic Routing Protocol Enhanced with Reinforcement Learning

We assume that packet exchanges occur opportunistically during node encounters, where nodes do not maintain end-to-end paths to destinations. Instead, they rely on a store-carry-forward mechanism, holding packets until they encounter a suitable relay or the destination node. For the routing process, each node employs a reinforcement learning model to optimize forwarding decisions based on past interactions with neighboring nodes. Using either Q-learning or PPO, nodes make forwarding decisions based on rewards derived from successful packet deliveries.

B. Epidemic Implementation in Mininet-Wifi

This section provides a detailed explanation of our implementation of the epidemic routing protocol. Inspired by the epidemic routing implementation in ns-3 [12], we have adapted it for use in Mininet-WiFi emulator. The epidemic routing protocol operates through a controlled flooding mechanism designed to reduce network overhead. Section III-B1 presents various parameters, that influence the behavior of epidemic routing. Section III-B2 describes the packet exchange process as nodes come within communication range. To incorporate Q-learning, each node maintains a Q-table, detailed in Section III-B3a. While if using PPO, nodes directly optimize the forwarding policy, detailed in Section III-B3b. Finally, Section

III-B4 covers the interaction and operation of these epidemic routing modules.

1) *Epidemic Parameters*: We use three parameters in our epidemic routing implementation, shown in Table I. First, the `HopCount` represents the maximum number of nodes that a packet can be transferred through before being discarded. This parameter should be equal to or greater than the number of network nodes to avoid discarding a packet before reaching the destination nodes. The `QueueLength` parameter shows the maximum number of packets that a queue can store. To prevent packet loss due to an overflow of queued packets, the value must be higher than the total number of unique packets in the emulation. The `BeaconInterval` specifies the interval, in seconds, at which a beacon packet is transmitted. This parameter was carefully chosen based on the rate at which packets are generated, to minimize delivery delay. For instance, if a new packet is created every five seconds, we should set the `BeaconInterval` to 5 seconds to allow the exchange of newly generated packets.

2) *Summary Vector Exchange*: In this section, we describe the summary vector exchange mechanism, which is how two nodes exchange their disjoint packets when they move in range of each other. The objective of the summary vector exchange is to avoid sending packets already present at the other node [6]. First, we assume that there are two nodes approaching each other in the same area, labeled A and B. Once these nodes come into range of one another, they start the summary vector exchange session as shown in Figure 2. Assuming node A finds that node B is within its range, node A will start the summary vector exchange session by sending summary vector A denoted as SV_A to node B. After receiving SV_A , B compares the packet IDs presented in A's summary vector with the packet IDs located in its own buffer, and determines the disjoint set of packets. This set is then sent back to node A. Finally, node B sends its summary vector to node A, which also determines the disjoint set of packets and sends them back to node B. At this point, both nodes have the same set of packets.

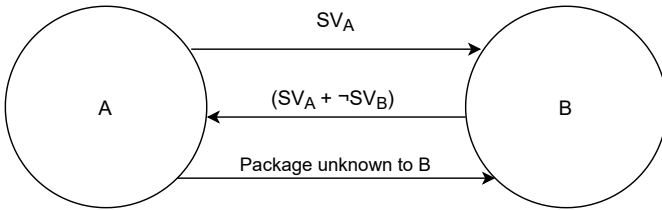


Fig. 2: The Epidemic Routing protocol when two nodes, A and B, come into transmission range of each other

3) *Reinforcement Learning Integration*: In this work, we extend the Epidemic Routing protocol by integrating a RL-based approach. The goal is to enhance decision-making during packet forwarding by dynamically selecting the most suitable neighbor for forwarding, reducing unnecessary transmissions, and optimizing packet delivery efficiency in OppNets.

a) *Q-Learning Algorithm*: We employ the Q-learning algorithm [13], which updates the Q-value for a given state-

action pair based on the observed reward and the maximum future expected reward. The Q-values are updated according to the following formula:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (1)$$

where:

- $Q(s, a)$ is the current Q-value for the state s and action a ,
- α is the learning rate, controlling how much new information overrides the old,
- r is the immediate reward received after taking action a ,
- γ is the discount factor, which controls the importance of future rewards,
- $\max_{a'} Q(s', a')$ is the maximum Q-value for the next state s' and possible actions a' .

In this algorithm, the node iteratively updates its Q-values based on the difference between the expected reward and the observed reward. Over time, this process enables the node to learn the optimal forwarding strategies based on the rewards received during interactions with neighboring nodes.

b) *Proximal Policy Optimization Algorithm*: PPO [14] uses a clipped surrogate objective function to limit the size of policy updates, which helps to avoid large, destabilizing updates. The objective function is defined as follows:

$$L^{\text{PPO}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (2)$$

where:

- $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ is the probability ratio between the new policy π_θ and the old policy $\pi_{\theta_{\text{old}}}$ for taking action a_t in state s_t ,
- $\hat{A}_t$ is the advantage estimate at time step t , representing the relative value of action a_t compared to the expected value of state s_t ,
- ϵ is a hyperparameter that controls the clipping range to limit how much the policy can change at each update.

In this algorithm, PPO alternates between collecting experience and optimizing the policy within the clipped objective function. The clipping mechanism ensures that updates stay within a predefined range, which improves learning stability. Over time, this iterative process enables nodes to learn an optimal forwarding policy by maximizing the cumulative rewards received during encounters with neighboring nodes.

c) *State, Action, and Reward Design*:

- **State s** : The state in this model is defined by the current node and the list of packet IDs waiting to be forwarded. In the implementation, we use the tuple (node name, packet IDs) to represent the state.
- **Action a** : The action corresponds to the selection of a neighboring node to which the packet will be forwarded. The action space consists of all neighboring nodes within range of the current node.
- **Reward r** : The reward function is designed to encourage efficient packet forwarding. If the packet is successfully

Parameter	Defaults	Summary
HopCount	10	Maximum number of times a packet will be flooded
QueueLength	64	Maximum number of packets that a queue can hold
BeaconInterval	10	Time in seconds after which a beacon packet is broadcast

TABLE I: Epidemic Routing Parameters

delivered, a positive reward of 1 is given. Otherwise, the reward is 0.

$$r = \begin{cases} 1 & \text{if all packets delivered successfully} \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

d) Implementation: Each node maintains a policy table or value function, where states (represented by the node and packet IDs) are mapped to the optimal actions (neighboring nodes) for forwarding packets. The node selects a neighbor based on its learned policy (using either Q-values in Q-learning or the policy function in PPO) and forwards the packet. After forwarding, it receives a reward by checking whether the packet reaches the chosen neighbor.

In the class implementing OppNet nodes with reinforcement learning algorithm, the following key functions enable the routing logic:

- `chooseNeighbor()`: Implements the epsilon-greedy strategy in Q-learning or a stochastic policy in PPO to select the next neighbor based on the current state.
- `updatePolicy()`: Updates the Q-value in Q-learning or adjusts the policy function in PPO based on rewards received.
- `getReward()`: Determines the reward by checking if the packet IDs are found in the chosen neighbor's buffer file, indicating a successful forwarding attempt.
- `sendSummaryVector()`: Manages the summary vector exchange, selects neighbors for forwarding, and updates the policy or Q-values accordingly.

The emulation is executed in the Mininet-WiFi environment, where each node operates its reinforcement learning agent and learns from interactions with neighboring nodes. Packet delivery performance is periodically evaluated, and metrics such as end-to-end delay, packet delivery ratio, and hop count are recorded. The learned policies allow nodes to improve their forwarding strategies over time, minimizing redundant transmissions and enhancing overall network efficiency.

4) Epidemic Routing with Reinforcement Learning Operation: In this section, we describe the operation of the Epidemic Routing protocol integrated with reinforcement learning algorithms, including Q-learning and PPO. The operation begins with each node periodically broadcasting a summary vector to its neighboring nodes. The process of epidemic routing with reinforcement learning is as follows:

- 1) **Initialization:** Each node initializes a `PacketQueue` to store buffered packets and sets up additional parameters discussed in Section III-B1. Using the `ListenSummaryVector` function, each node contin-

uously listens to summary vector requests received from other nodes in the network.

- 2) **Summary Vector Broadcast:** Each node generates and broadcasts its summary vector, which contains the packet IDs of the packets it holds. This is done using the method `sendSummaryVector`, where the summary vector includes information about the packets stored in the node's buffer, such as packet IDs, source and destination addresses, hop count, and packet content, which is then sent to neighboring nodes.
- 3) **Neighbor Selection Based on Policy:** After broadcasting the summary vector, the node must decide which neighbor to forward the packet to, based on its learned policy. For Q-learning, this selection uses an ϵ -greedy strategy in `choose_neighbor`, where the node either explores a random neighbor (with probability ϵ) or exploits its learned Q-values (with probability $1 - \epsilon$). For PPO, the node selects the neighbor based on a learned policy function that considers past rewards and encounters to maximize the expected reward.
- 4) **Policy or Value Update:** Following packet forwarding, the node waits for feedback from the neighbor. If the packet is successfully received by the neighbor, the node receives a reward. The reward function, implemented in `get_reward`, checks if the packet has been found in the neighbor's buffer. The node then updates its policy based on this feedback, using either the Q-learning update rule as defined in III-B3a or PPO's policy gradient adjustment defined in III-B3b.
- 5) **Packet Transfer:** Finally, the node sends the selected packets to the chosen neighbor via a peer-to-peer connection. For our emulation, this is achieved by executing a command through the shell using `nc` (netcat) to simulate packet transfer.

This integration of reinforcement learning algorithms, such as Q-learning and PPO, with Epidemic Routing allows nodes to intelligently choose neighbors based on past experiences, which improves packet forwarding efficiency and reduces redundant transmissions. Over time, nodes learn to prioritize neighbors that maximize successful packet delivery while minimizing end-to-end delay and hop count.

The full implementation is accessible on GitHub.²

IV. PERFORMANCE EVALUATION

A. Experiment setup

We perform the emulations over an area of 1000×300 m². All the emulations are averaged over 100 runs with each

²<https://github.com/HuyPHD2024-2027/mininet-wifi>

Parameter	Value
Area	1000 m × 300 m
Number of runs	100
Emulation time	200,000 s
Number of network nodes	50
Number of packets	2450
Packet generating time	10
Propagation loss model	logDistance
Mobility model	Random Direction
Node speed	0-20 m/s

TABLE II: Emulation Parameters

emulation running for 200,000 s. Emulations are performed with 50 nodes and each node acts a packet generator. Each of the 50 nodes sends one packet to the other 49 nodes with a total of 2450 packets at 10 s emulation time. Propagation loss model *logDistance* is used to limit the transmission ranges of the nodes. The transmission range of the nodes is set as 100 m for evaluation. The mobility model used is Random Direction [15] with seed at 0.01 (20.0 m/s). A summary of all emulation parameters is shown in Table I. Furthermore, a summary of the Epidemic routing protocol parameters is shown in Table II.

B. Results and Analysis

In this section, we present the performance evaluation of the proposed Epidemic Routing protocol enhanced with two RL algorithms Q-learning and PPO, compared to the Traditional Epidemic Routing protocol [12]. The emulation results are analyzed in terms of the impact of network size on packet delivery latency, routing overhead, and number of hops. Figures 3, 4, and 5 demonstrate these metrics.

1) *Packet Delivery Latency*: Figure 3 presents the packet delivery ratio between the proposed Epidemic Routing protocol with Q-learning, PPO and the Traditional Epidemic Routing protocol as the packet delivery latency increases. The results indicate that at extreme latency conditions or when the latency is constrained to a very small duration (below 2 seconds with 5% of packet delivered) and when it reaches a very large duration (around 1000 seconds and nearly 100% of packet delivered), the difference in packet delivery ratio between the three protocols is minimal. However, as the delivery latency grows from 10 to 200 seconds, the packet delivery ratio improves for all three protocols, with the Q-learning-enhanced protocol consistently achieving the lowest latency, followed by the PPO-based protocol, and then the Traditional Epidemic Routing protocol.

This observed improvement can be attributed to the adaptive decision-making capabilities of the Q-learning algorithm, which enables nodes to identify and select more optimal neighbors for packet forwarding. By learning from previous transmissions, the Q-learning mechanism minimizes redundant transmissions and avoids network congestion, thereby reducing delivery latency. PPO with policy-based approach requires more extensive

computational resources and a larger amount of data for each policy update. As a result, PPO typically incurs a longer decision-making time than Q-learning, as it must process a more complex action space and perform multiple optimization steps before selecting an optimal forwarding action. This makes PPO slower than Q-learning when reaching a forwarding decision, though it remains effective in environments that demand adaptability to continuous changes. Meanwhile, the Traditional Epidemic Routing protocol relies on indiscriminate packet broadcasting, which results in higher network congestion, leading to increased delivery latency, particularly as the network size and traffic volume grow. Consequently, the Q-learning approach offers a more efficient routing strategy with latency constraints.

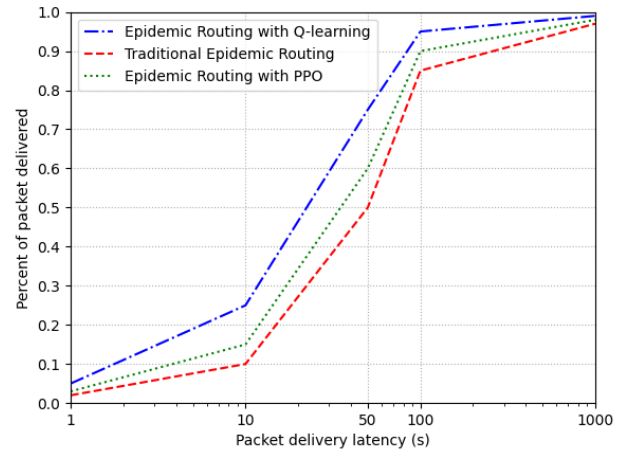


Fig. 3: Impact of Network Size on Packet Delivery Latency

2) *Routing Overhead*: Figure 4 illustrates the impact of network size on routing overhead, measured in *kb/s*. As the number of nodes in the network increases, the overhead grows for all three protocols: Traditional Epidemic Routing, Epidemic Routing with Q-learning, and Epidemic Routing with PPO. However, they differ in the amount of overhead generated as number of network nodes increases.

The Traditional Epidemic Routing protocol has the highest overhead, reaching approximately 800 kbps at 50 nodes, because it broadcasts to all neighboring nodes within communication range. This method leads to substantial resource consumption and network congestion due to redundant transmissions, especially as the network becomes denser.

In contrast, the Epidemic Routing protocol with Q-learning reduces overhead by learning optimal forwarding paths, achieving around 420 kbps at 50 nodes. This approach allows nodes to select neighbors that maximize successful delivery while minimizing redundant transmissions. Consequently, Q-learning enhances efficiency, particularly in moderate to dense networks.

The Epidemic Routing protocol with PPO has the lowest routing overhead, around 300 kbps at 50 nodes. Although PPO requires more computational resources and time for each decision than Q-learning, it proves effective in minimizing

overhead by continuously refining its forwarding policy.

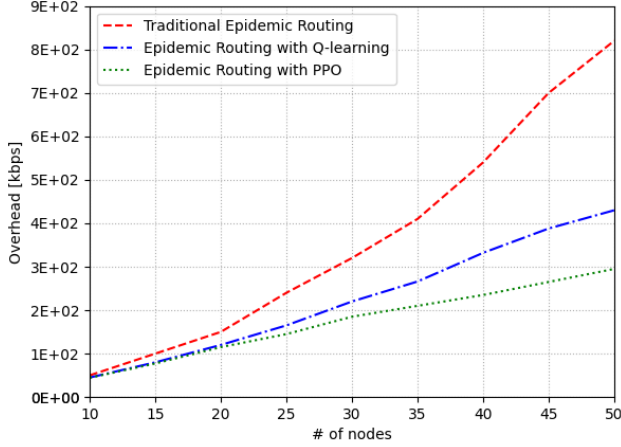


Fig. 4: Impact of Network Size on Routing Overhead

3) *Number of Hops*: Figure 5 illustrates the effect of increasing the number of nodes on number of hops that packets has to be transferred through. When network size is small (from 5 to 15 nodes) the difference in success rate is insignificant (around 2 and 4 respectively). However, the PPO approach consistently outperforms the Q-learning and the Traditional Epidemic Routing protocols when number of nodes in network increases. For instance, when network size reaches 50 nodes, The Epidemic Routing protocol with PPO achieves the lowest hop count, approximately 5 hops at 50 nodes. Epidemic Routing protocol with Q-learning requires 6 hops while Traditional Epidemic Routing needs up to around 9 hops.

As the network size increases, the PPO-enhanced protocol achieves the lowest hop count, followed by the Q-learning-enhanced protocol, while the Traditional Epidemic Routing protocol incurs the highest hop count. This comparison highlights the adaptability and efficiency of reinforcement learning approaches, especially in managing hop count in larger, dynamic opportunistic networks.

V. CONCLUSION

In this paper, we proposed an enhancement to the Epidemic Routing protocol by integrating Q-learning and PPO, which significantly improves the efficiency of packet forwarding in OppNets. This improvement is achieved by enabling nodes to intelligently select optimal forwarding paths, minimizing redundant transmissions and resource consumption. The results demonstrate that approach with Reinforcement Learning algorithms are well-suited for networks with constrained computational and storage resources, offering a scalable solution for delay-tolerant networking environments. As future work, we plan to integrate blockchain and Conflict-free Replicated Data Types to further ensure consistency and enhance the reliability of forwarding decisions, thus improving trustworthiness in highly dynamic and resource-limited OppNets.

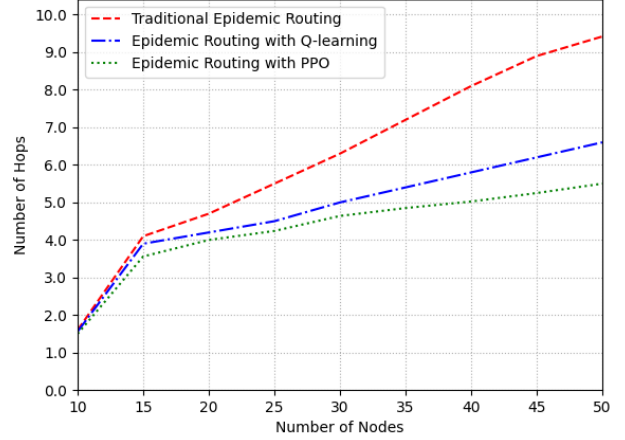


Fig. 5: Impact of Network Size on Number of Hops

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